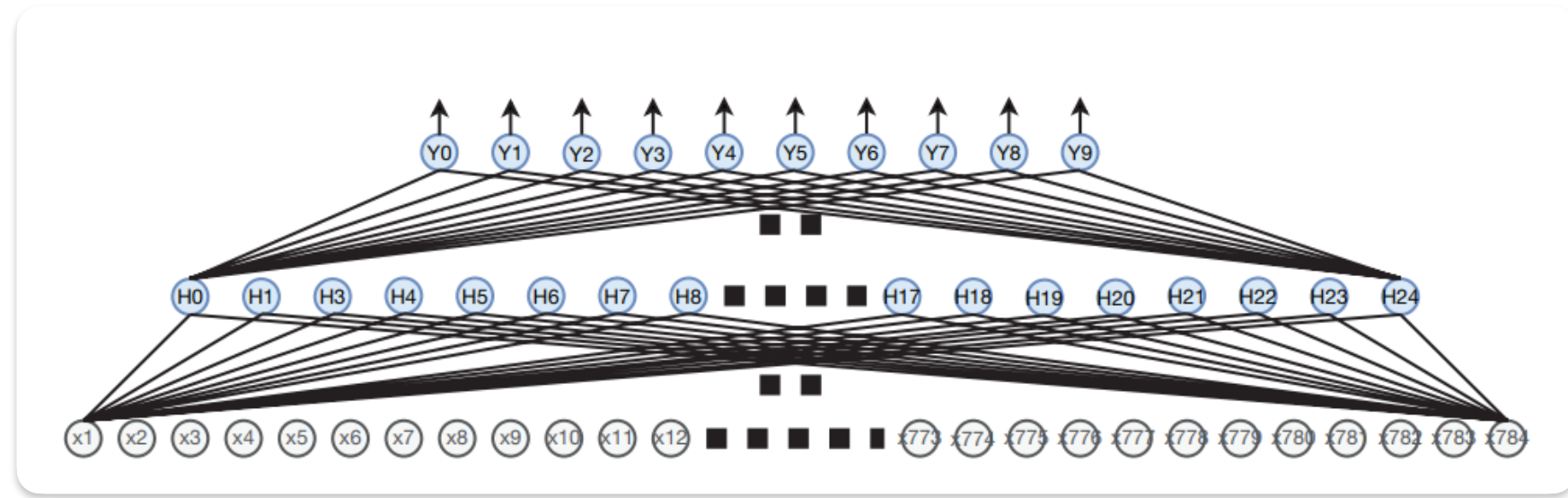


1. Spatial Relationship in Neural Networks:

A.



- B. the figure above is a fully connected neural network which transfer a 2D grid (28×28 pixels) into 1D vector (784 pixels), all 25 hidden perceptrons see all 784 pixels, so all pixels are equally close to each other from the perspective of a single perceptron. it appears that info related to which pixels are neighboring each other is lost.
- C. if it is beneficial to take spatial relationship between pixels into account, then neural networks have to learn it by themselves. convolutional neural networks are able to take each pixel location into account.